

Introduction

Grouped count data arise naturally in many applied settings. In consumer behaviour research, households make repeated purchases across a large number of product categories; in agriculture, field plots are observed repeatedly across seasons with counts recorded for different seed varieties. In both cases, the data share a common structure. Observations are grouped, counts are recorded across Q categories and within-group dependence is an intrinsic feature of the data.

The multinomial distribution provides a natural model for such data, but direct estimation is computationally prohibitive when Q is large. Taddy (2015) addressed this through the Distributed Multinomial Regression (DMR) framework, which exploits the exact Poisson–Multinomial factorisation

$$\prod_{q=1}^Q P(z_q; \mu_q) = P(S; \sum_q \mu_q) \cdot M(z; p, S) \quad (1)$$

to decompose the multinomial regression into Q independent Poisson regressions that can be solved in parallel. Lee, Green and Ryan (2017) extended DMR to grouped data by introducing group-level random effects λ_j , accommodating within-group dependence while retaining the Poisson surrogate structure. However, their model assumes λ_{jq} to be independent across categories q , which rules out the cross-category correlation that is pervasive in high-dimensional settings.

Model Specification

Data Structure and Notation

Setup.

- $j = 1, 2, \dots, J$: index of groups
- $i = 1, 2, \dots, n_j$: index of observation within group j
- Q : number of “classification” corresponding to the response Q -dimensional count vector $Y_{ij} = (Y_{ij1}, \dots, Y_{ijQ})^T \in \mathbb{Z}_{\geq 0}^Q$, where Y_{ijq} denotes the count of category q
- p : number of covariates
- v_{ij} : covariate vector, where $v_{ij} \in \mathbb{R}^p$
- δ_{ij} : known positive scalar offset

and define the following quantities:

$$Y_{ij+} = \sum_{q=1}^Q Y_{ijq} \quad Y_{+jq} = \sum_{i=1}^{n_j} Y_{ijq} \quad Y_{j++} = \sum_{i=1}^{n_j} \sum_{q=1}^Q Y_{ijq} \quad (2)$$

denoting the observation-level total, the group-category total and the group-level grand total respectively. The total number of observations across all groups is $N = \sum_{j=1}^J n_j$.

The Poisson-Multinomial Equivalence

Lemma 1 (Poisson-Multinomial Factorisation). Let $Z_1, \dots, Z_Q$ be independent with $Z_q \sim P(\mu_q)$ for $q = 1, 2, \dots, Q$, where $\mu_q > 0$. Define $S = \sum_{q=1}^Q Z_q$ and the probability vector $p = (\frac{\mu_1}{\mu_+}, \dots, \frac{\mu_Q}{\mu_+})^T$, where $\mu_+ = \sum_{q=1}^Q \mu_q$. Then the joint distribution of $(Z_1, \dots, Z_Q)$ factorises exactly as:

$$p(Z_1 = z_1, \dots, Z_Q = z_Q) = P(S = s; \mu_+) \cdot M(z; p, s) \quad (3)$$

where $s = \sum_{q=1}^Q z_q$, $P(\sim; \mu_+)$ denotes the Poisson probability mass function with poisson parameter μ_+ and $M(\sim; p, s)$ denotes the multinomial probability mass function given total s and probability vector p .

Proof. Since $Z_1, \dots, Z_Q$ are independent with $Z_q \sim P(\mu_q)$, by direct computation:

$$\prod_{q=1}^Q \frac{e^{-\mu_q} \mu_q^{z_q}}{z_q!} = e^{-\mu_+} \cdot \frac{\mu_+^s}{s!} \cdot \frac{s!}{\prod_{q=1}^Q z_q!} \prod_{q=1}^Q \left(\frac{\mu_q}{\mu_+} \right)^{z_q} = P(s; \mu_+) \cdot M(z; p, s). \quad (4)$$

□

This factorisation is exact and holds for all $(z_1, \dots, z_Q) \in \mathbb{Z}_{\geq 0}^Q$.

Remark 1. Lemma has an important implication for inference.

Conditional Model Given Random Effects

To discuss within-group dependence, we introduce a Q -dimensional vector of group-specific random effects $\lambda_j = (\lambda_{j1}, \dots, \lambda_{jQ})^T$, where $\lambda_{jq} > 0$ represents the multiplicative deviation of group j 's baseline intensity for category q from the population average. Conditional on λ_j , we assume that all counts within group j are independent:

$$Y_{ijq} | \lambda_j \sim P(\delta_{ij} \lambda_{jq} \zeta_{ijq}), i = i, \dots, n_j, q = 1, \dots, Q \quad (5)$$

where the fixed-effect component of the conditional mean is:

$$\zeta_{ijq} = \exp(v_{ij}^T \phi_q) \quad (6)$$

and $\phi_q \in \mathbb{R}^p$ is a vector of regression coefficients for category q . The random effect λ_{jq} captures the persistent tendency of group j toward category q that is not explained by the observed covariates v_{ij} . Observations within the same group share one same λ_j , which induces within-group dependence even after conditioning on covariates.

Remark 2 (Conditional Poisson-Multinomial Equivalence). By Lemma 1 applied conditionally, model 5 implies that:

$$Y_{ij+} | Y_{ij+}, \lambda_j \sim M(Y_{ij+}, p_{ij}), \text{ where } p_{ijq} = \frac{\lambda_{jq} \zeta_{ijq}}{\sum_{q=1}^Q \lambda_{jq} \zeta_{ijq}} \quad (7)$$

and simultaneously:

$$Y_{ij+} | \lambda_j \sim P(\delta_{ij} \sum_{q=1}^Q \lambda_{jq} \zeta_{ijq}). \quad (8)$$

The Poisson–Multinomial equivalence therefore holds exactly at the conditional level, given λ_j . The multinomial probability vector p_{ij} depends on λ_j only through the ratios $\frac{\lambda_{jq}}{\sum_q \lambda_{jq}}$, so the random effects act as multiplicative modulation of the category probabilities that varies across groups but remains constant within a group across observations.

Factor-Structured Random Effects

The cross-category correlation is driven by common latent factors rather than being independently determined for each category. We propose a factor-structured random effects model of λ_j across groups. Under the independence assumption of Lee et al.(2017), λ_{jq} are independent across q , so the cross-category covariance of λ_j is diagonal. This assumption becomes increasingly restrictive as Q grows, since it rules out the systematic co-variation of preferences across categories that is typical in high-dimensional count data. We relax this assumption by modelling the log-transformed random effects through a latent factor model:

$$\log \lambda_j = \mu + B f_j + \epsilon_j \quad (9)$$

where the components are defined as follows. The vector $\mu = (\mu_1, \dots, \mu_Q)^T \in \mathbb{R}^Q$ denotes the population-level baseline log-intensities for each category, the matrix $B = [b_1, \dots, b_K] \in \mathbb{R}^{Q \times K}$ is the factor loading matrix with $K \ll Q$ and its entries B_{qk} denotes the sensitivity of category q 's random effect to the k -th latent factor. The vector $f_j \in \mathbb{R}^K$ denotes the group-specific latent factor scores assumed as follows, distributed identically and independently:

$$f_j \sim N_K(0, I_K) \quad (10)$$

The vector $\epsilon_j \in \mathbb{R}^Q$ denotes the error term capturing category-specific variation in λ_j that is not attributable to the K common factors:

$$\epsilon_j \sim N_Q(0, D), \quad D = \text{diag}(d_1^2, \dots, d_Q^2) \quad (11)$$

with $f_j \perp \epsilon_j$ for all j . Then the marginal distribution of $\log \lambda_j$ is:

$$\log \lambda_j \sim N_Q(\mu, \Sigma), \quad \Sigma = BB^T + D \quad (12)$$